Predicting Stroke With Accessible Machine Learning Software Application

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Advanced Java Topics & Machine Learning
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Introduction

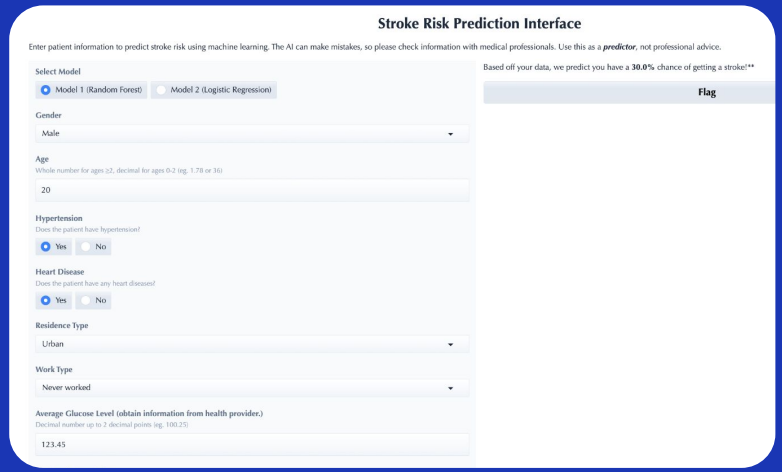
- Stroke is a leading cause of death and serious long-term disability, causing about 1 in 6 cardiovascular-related deaths.
- Someone in the world has a stroke every 40 seconds, and someone dies of a stroke every 3 minutes and 14 seconds.
- **1 in 4 adults over age 25 will have a stroke in their lifetime**.

My App

My app allows people of any age group who possess their basic health information to **predict their likeliness of getting a stroke in the future.**

This can help people understand their need to seek further professional medical advice, receive necessary steps to prevent it, and **intervene early.**

Early identification and intervention of stroke has been **statistically proven to reduce mortality rates.**



Data Source

Stroke Prediction Dataset

11 clinical features for predicting stroke events

Last Updated: 5 years ago (Version 1)

healthcare-dataset-stroke-data.csv (316.97 kB)

Detail Compact Column

About this file

The data contains 5110 observations with 12 attributes.

gender Gender	age Age	hypertension Hypertension binary feature	heart_disease Heart disease binary feature	ever_married Has the patient ever been married?	work_type Work type of the patient
Female 59% Male 41% Other (1) 0%	0.08 – 82	0 – 1	0 – 1	true 3353 66% false 1757 34%	Private 57% Self-employed 16% Other (1366) 27%
Male	67	0	1	Yes	Private
Female	61	0	0	Yes	Self-employed
Male	80	0	1	Yes	Private
Female	49	0	0	Yes	Private
Female	79	1	0	Yes	Self-employed
Male	81	0	0	Yes	Private

5110 entries, 11 clinical attributes for predicting stroke (represented by '1' for stroke, '0' for no stroke).

Training Variables

- Gender
- Age
- Hypertension presence (Y/N)
- Heart disease presence (Y/N)
- Work type
- Residence type (Urban/Rural)
- Average glucose level
- BMI

Data Workflow Overview

Data Cleaning

Encoding Categorical Variables:

```
gender =
'Male':0,'Female':1,'Other':-1
Residence_type = 'Rural':0,'Urban':1
work_type =
'Private':0,'Self-employed':1,'Govt_job':2,'children':-1,'Never_worked':-2
```

SMOTE
Data Augmentation

Model
Training

RandomForestClassifier

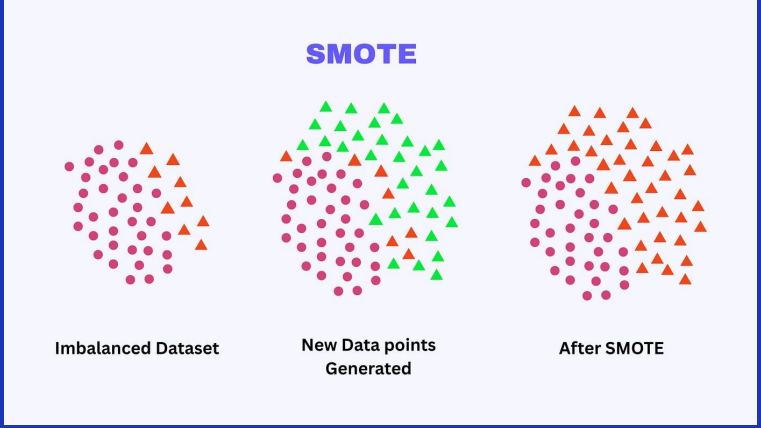
Logistic Regression

Testing &
Evaluation

Algorithm: Synthetic Minority Oversampling Technique (SMOTE)

- SMOTE is a technique used for addressing class imbalance by creating synthetic samples for the minority class. It interpolates new data points between existing minority samples and their nearest neighbors.

- Often reduces overfitting and improves classification performance on imbalanced datasets.

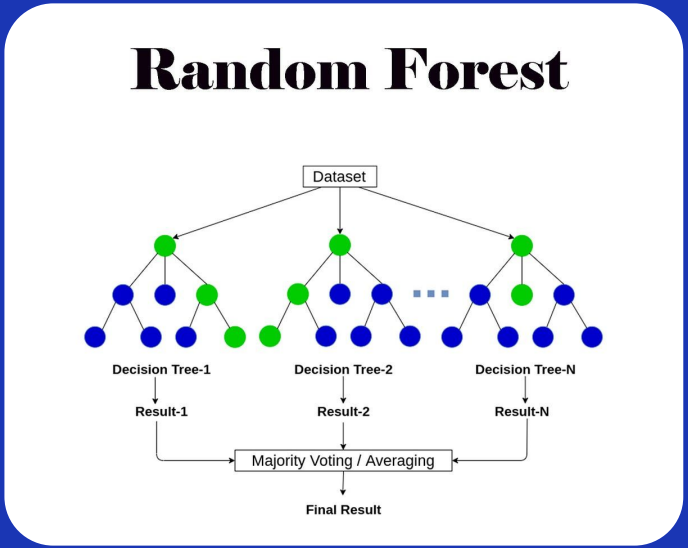


Algorithm: Random Forest

- Random Forest is an ensemble machine learning algorithm used for both classification and regression.

- Creates a “forest” of decision trees, and uses bootstrapping (random sampling) to create several random subsets of the training data for each tree

- Each tree makes its own individual prediction based on the data it was trained on, and the algorithm aggregates a result

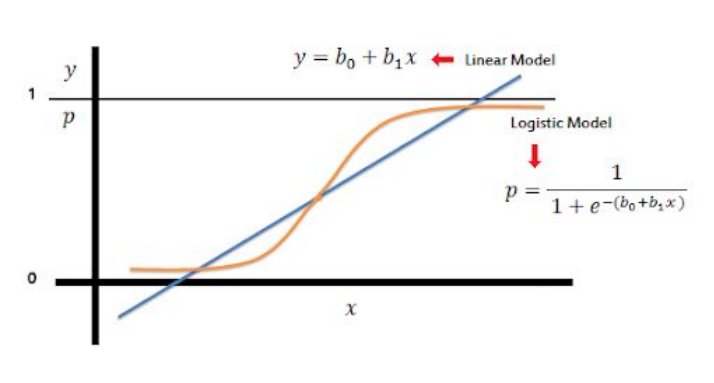


Algorithm: Logistic Regression

- Logistic Regression is a supervised machine learning algorithm for binary classification. It models the probability of an event using a sigmoid function.

- Computes a linear combination of input features and their weights plus a bias term to produce logit score.

- Passes the logit score into the sigmoid function to produce a probability.



Program Pros & Cons

With a user-friendly app using high-accuracy AI models that may be widely distributed, more people can be prepared to encounter a stroke and be informed of when to receive care.

However, the app holds some limitations:

- It is not explainable
- The AI cannot provide professional medical advice (next steps, what to look out for in the future, etc)
- Models are trained on limited data
  - Not perfectly accurate
  - May contain unnoticed biases

Stroke Risk Prediction Interface

Enter patient information to predict stroke risk using machine learning. The AI can make mistakes, so please check information with medical professionals. Use this as a ***predictor***, not professional advice.

Important Warnings:

⚠️ **Critically high glucose level.** Please consider seeking medical attention immediately.

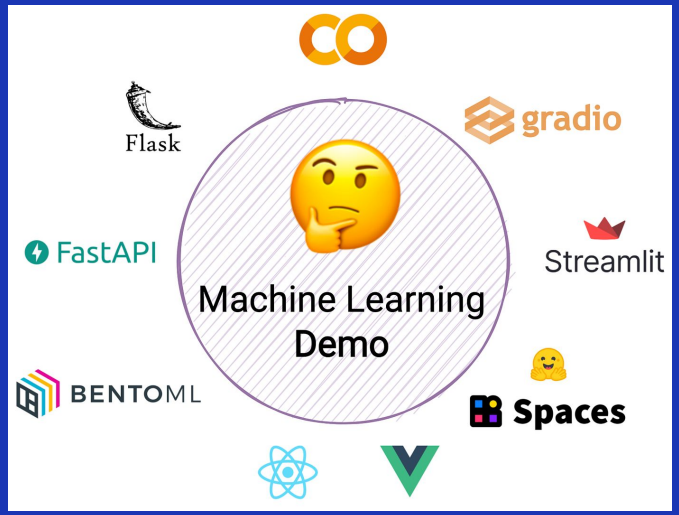
⚠️ **Pediatric case**: Model primarily trained on adult data. Use with caution.

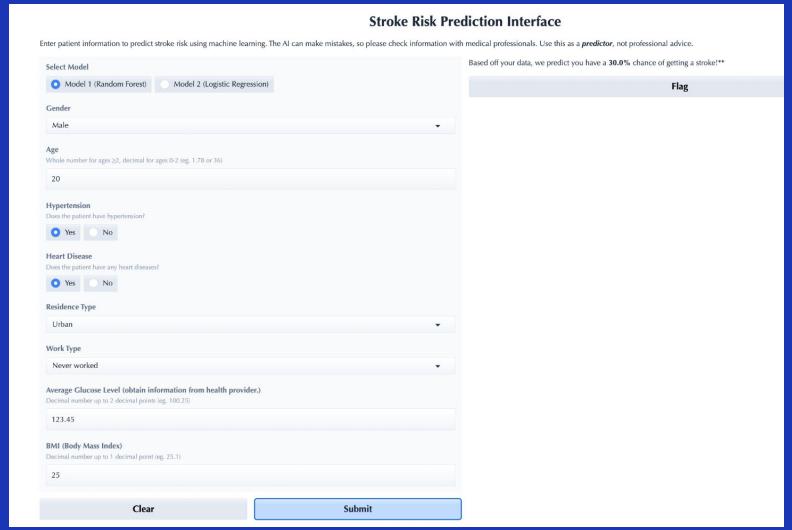
⚠️ **Limited pediatric obesity data in training set. (high BMI & age <18)**

Clinical Review Recommended: Your statistics suggest that the prediction should be reviewed by a healthcare professional, as the model may be inaccurate. (You might be getting this warning because you are below 18, your average glucose level is >200, or your BMI is above 35.)

Challenges

- Difficulty in finding ways to address class imbalance
- Difficulty in implementing the Gradio UI, exporting ML models to client class



Conclusion/Takeaways

- Learned about machine learning demo UI implementation with Gradio
- Learned more about how Random Forest and Logistic Regression models work on the inside
- Explored ways to address the limitations of AI technology for healthcare